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MIL-STD-2045-17505
29 July 1994

MILITARY STANDARD

Information Technology
DOD Standardized Profile

Internet Domain Name System (DNS)

AMSC N/A

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Foreword

This military standard is approved for use by all Departments and Agencies of the Department of Defense (DOD).

Beneficial comments (recommendations, additions, deletions) and any pertinent data that may be of use in improving this MIL-STD should be addressed to the:

Joint Interoperability and Engineering Organization (JIEO)
ATTN: JEBBF
Fort Monmouth, New Jersey 07703-5613

by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this MIL-STD or by memorandum.

This MIL-STD 2045-17505 DOD Standardized Profile (DSP) is a functional standard produced by the Data Communications Protocol Standards (DCPS) Technical Management Panel (DTMP). DTMP functional standards are functional groupings of base standards. Referenced base standards may be commercial, DOD or de facto standards, although International Standards (produced by ISO, CCITT, and other bodies) are preferred when possible.

This Defense Standardized Profile (DSP) is a functional DOD Data Communications Protocol Standard (DCPS) produced by the DCPS Technical Management Panel (DTMP). The MIL-STD-2045 document series was established within the DCPS Standardization Area to allow for the enhancement of commercial standards or the development of standards that are unique to DOD.

The MIL-STD-2045-10000 series, MIL-STD-2045-10000 to MIL-STD-2045-19999 inclusive, will be used to describe how DOD will implement commercial, international, national, federal, or military standards within the functional profile concept to provide required network services. The Government Open Systems Interconnection Profiles (GOSIP) will serve as the base for developing the 10000 series with DOD enhancements, unique military standards, and interim standards being used only when necessary.

The MIL-STD-2045-20000 series, MIL-STD-2045-20000 to MIL-STD-2045-29999 inclusive, will be used to describe DOD enhancements and extensions to existing commercial, international, national, or federal standards.

The MIL-STD-2045-30000 series, MIL-STD-2045-30000 to MIL-STD-2045-39999 inclusive, will be used to describe protocols and services unique to DOD that will not be supported by commercial, international, national, or federal standards.

The MIL-STD-2045-40000 series, MIL-STD-2045-40000 to MIL-STD-2045-49999 inclusive, will be used to document interim standards. Interim standards document protocols and services needed by DOD until these protocols and services are described in either a GOSIP or a MIL-STD-2045-20000 or -30000 series standard.

The MIL-STD-2045-50000 series, MIL-STD-2045-50000 to MIL-STD-2045-59999 inclusive, will be used to describe how DOD will implement commercial, international, national, federal, or military standards within the functional profile concept to provide required network services. The Government Open Systems Interconnection Profiles (GOSIP) will serve as the base for developing the 50000 series with DOD enhancements, unique military standards, and interim standards being used only when necessary. The difference between MIL-STD-2045-10000 series and the MIL-STD-2045-50000 series is that the 50000 series are interim profiles

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Specific details and instructions for establishing a MIL-STD-2045 document, as well as profile development guidelines, are documented in MIL-HDBK-829. DTMP Working Groups shall be responsible for DSP development and informal Service or Agency coordination; the DTMP Plenary shall be responsible for final review and approval.

This document is part of a set of interim DOD data communications protocol profiles based on the Internet protocol suite and is intended to support the interoperability of DOD communication networks, including connectivity with the Defense Data Network (DDN).

This part of MIL-STD 2045-17505 contains one normative annex and one informative annex:

Annex A (normative)	DSPICS Requirements List (DPRL).
Annex B (information)	Concluding Material.

For DOD acquisition purposes, where such differences exist, this DSP shall be the controlling document.

The Preparing Activity for this standard is the Data Communication Protocol Standards Technical Management Panel (DTMP). The custodians for the document are identified in the Defense Standardization Program, "Standardization Directory (SD-1)" and are classified in the Federal Supply Classification (FSC) system under Data Communication Protocol Standards (DCPS). Additional information can be obtained from:

Joint Interoperability and Engineering Organization
ATTN: DTMP Chairman
Ft. Monmouth, New Jersey 07703-5613

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Introduction

This DOD Standardized Profile (DSP) is defined within the context of functional standardization, in accordance with the principles specified by ISO/IEC TR 10000, "Framework and Taxonomy of International Standardized Profiles" and MIL-HDBK-829. The context of functional standardization is one part of the overall field of Information Technology (IT) standardization activities - covering base standards, profiles, and registration mechanisms. A profile defines a combination of base standards that collectively perform a specific well-defined IT function. Profiles standardize the use of options and other variations in the base standards to promote system interoperability and to provide a basis for the development of uniform, internationally recognized system tests.

One of the most important roles for a DSP is to serve as the basis for the development of recognized tests. DSPs also guide implementors in developing systems that fit the needs of the US Department of Defense (DoD). DSPs are produced not simply to 'legitimize' a particular choice of base standards and options, but to promote real system interoperability. The development and widespread acceptance of tests based on this and other DSPs is crucial to the successful realization of this goal.

The base standards of this MIL-STD include Request For Comments (RFCs) designated as Official Internet Architecture Board (IAB) standards, and other RFCs.

The specifications in this part of DOD Standardized Profile (DSP) 2045-17505 cover the Domain Name System. The current technical content of this document has been derived wherever possible from Internet Architecture Board (IAB) Standard (STD) 13. This document must be combined with IAB STD 13 (RFC 1034: November 1987, Domain Names - Concepts and Facilities, RFC 1035: November 1987, Domain Names - Implementation and Specification) and IAB STD 3 (RFC 1123: October 1989, Requirements For Internet Hosts– Application and Support).

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Information Technology - DOD Standardized Profile - Internet Domain Name System (DNS)

1 Scope

1.1 General

This DOD Standardized Profile (DSP) applies to the Domain Name System (DNS).

1.2 Position within the taxonomy

This profile is classified as MIL-STD 2045-17505 in accordance with MIL-HDBK 829.

1.3 Scenario

This MIL-STD specifies the provisions of the Internet Domain Name System (DNS) Profile for DOD Communications.

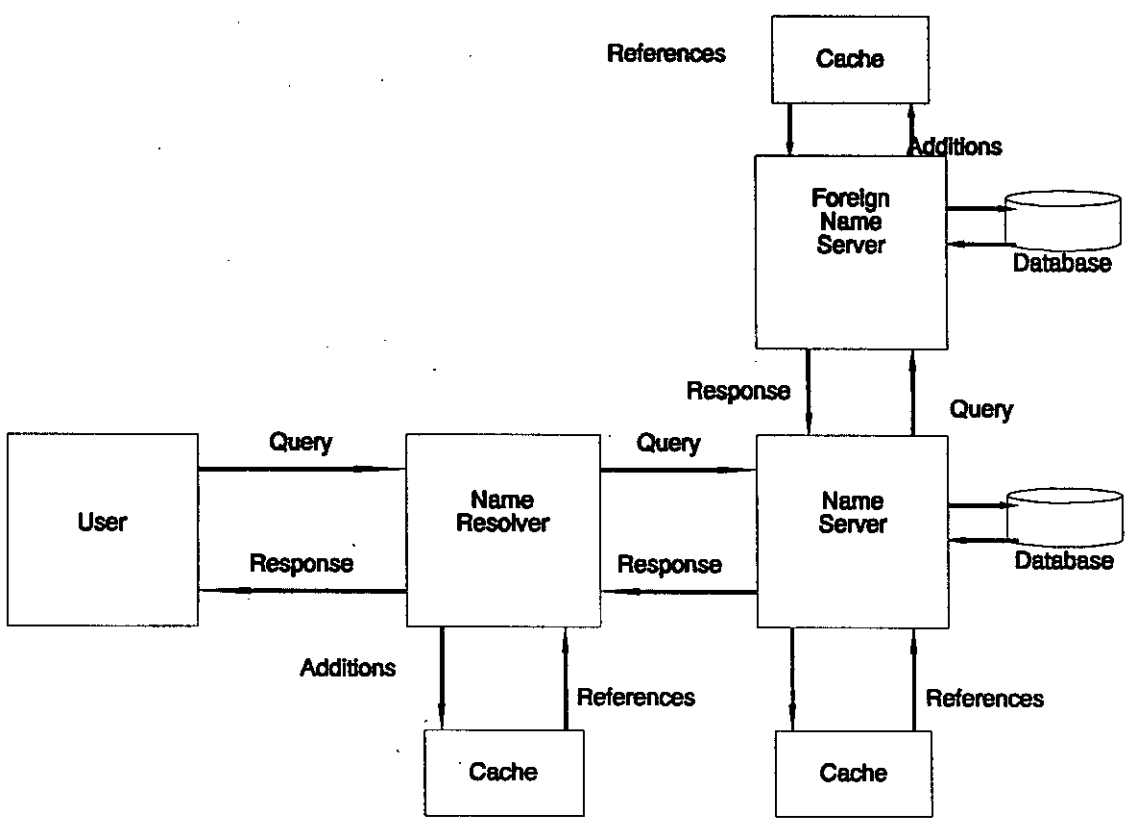


Figure 1 DNS Scenario

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2 References

The following documents contain provisions which, through reference in this text, constitute provision of this part of MIL-STD 2045-17505. At the time of publication, the editions indicated were valid. All documents are subject to revision, and parties to agreements based on this part of MIL-STD 2045-17505 are warned against automatically applying any more recent editions of the documents listed below, since the nature of references made by MIL-STDs to such documents is that they may be specific to a particular edition.

2.1 Government Documents

2.1.1 Specifications, Standards, and Handbooks

The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

MIL-HDBK 829: July 1994 *Guidelines for DOD Standardized Profiles.*

DOD activities may obtain copies of DOD directives through their own publication channels or from the DOD Single Stock Point, Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094. Other federal agencies and the public may purchase copies from the U.S. Department of Commerce, National Technical Information Service, 5285 Port Royal Road, Springfield, VA 22161.

Copies of Federal Information Processing Standards (FIPS) are available to Department of Defense activities from the Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120-5099. Others must request copies of FIPS from the National Technical Information Services, 5285 Port Royal, Springfield, VA 22161-2171.

2.1.2 Other Government Documents, Drawings, and Publications

The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

None

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation.

2.2.1 Profiles

None

2.2.2 Base Standards

IAB STD 3 *(RFC 1123: October 1989, Requirements for Internet Hosts -- Application and Support)*

IAB STD 13 *(RFC 1034: November 1987, Domain Names - Concepts and Facilities, RFC 1035: November 1987, Domain Names - Implementation and Specification)*

2.2.3 Other Non-Government documents, drawings, and publications

The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation.

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IAB STD 6 *(RFC 768: August 1980, User Datagram Protocol).*

IAB STD 7 *(RFC 793: 1981, Transmission Control Protocol).*

IAB STD 14 *(RFC 974: January 1986, Mail Routing and the Domain System)*

RFC 952: *October 1985, DOD Internet Host Table Specification*

RFC 953: *October 1985, Host Name Server*

RFC 1032: *November 1987, Domain Administrators Guide*

RFC 1033: *November 1987, Domain Administrators Guide*

Internet Protocol Handbook, Vol.4, NIC 50007: *August 1989, The Domain Name System Handbook*

RFC's are public domain and are available on the Internet.

2.3 Order of precedence

In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3 Definitions

Internet Architecture Board (IAB) Standards (STD): The IAB has established this as an official standard protocol for the Internet. These protocols are assigned STD numbers.

Request For Comments (RFCs): RFCs are the working notes of the "Network Working Group," that is the Internet research and development community.

Note: All standards are published as RFCs, but not all RFCs specify standards.

4 Abbreviations and Acronyms

A	Address
CNAME	Canonical Name
EMAILBX	Error Mailbox
HINFO	Host Information
IAB	Internet Architecture Board
MB	Mailbox
MD	Mail Delivery
MF	Mail Forwarding
MG	Mail Group
MR	Mail Rename
MX	Mail Exchange
PTR	Pointer
QTYPE	Query Type
QCLASS	Query Class
RFC	Request For Comments
RMAILBX	Responsible Mailbox
RR	Resource Record

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SOA	Start of Zone Authority
STD	Standard
TCP	Transmission Control Protocol
TTL	Time to Live
TXT	Text
UDP	User Datagram Protocol
WKS	Well Known Service

5 Requirements

5.1 General Requirements

Implementations claiming conformance to this part of MIL-STD 2045-17505 shall support the following procedures as stated.

A conforming implementation of this profile shall be unconditionally compliant and therefore, shall satisfy all the "MUST" and all the "SHOULD" requirements of the reference base standards and shall not implement any capability that has been identified by the base standards as "SHOULD NOT".

5.1.1 Conventions

There are no additional requirements as specified in RFC 1035, section 2.3.

5.1.1.1 Preferred name syntax

There are no additional requirements as specified in RFC 1035, section 2.3.1 and as amended by RFC 1123, section 6.1.3.5.

5.1.1.2 Data Transmission Order

There are no additional requirements as specified in RFC 1035, section 2.3.2.

5.1.1.3 Character Case

There are no additional requirements as specified in RFC 1035, section 2.3.3.

5.1.1.4 Size Limits

There are no additional requirements as specified in RFC 1035, section 2.3.4.

5.2 Conformance Requirements

5.2.1 Domain Name Space and RR Definitions

5.2.1.1 Name Space definitions

There are no additional requirements as specified in RFC 1035, section 3.1.

5.2.2 Resource Record (RR) definitions

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5.2.2.1 Format

There are no additional requirements as specified in RFC 1035, section 3.2.1 and as amended by RFC 1123, section 6.1.2.1.

5.2.2.2 TYPE values

There are no additional requirements as specified in RFC 1035, section 3.2.2.

5.2.2.3 QTYPE values

There are no additional requirements as specified in RFC 1035, section 3.2.3.

5.2.2.4 CLASS values

There are no additional requirements as specified in RFC 1035, section 3.2.4.

5.2.2.5 QCLASS values

There are no additional requirements as specified in RFC 1035, section 3.2.5 and as amended by RFC 1123, section 6.1.2.2.

5.3 Standard RRs

5.3.1 CNAME RDATA format

There are no additional requirements as specified in RFC 1035, section 3.3.1.

5.3.2 HINFO RDATA format

There are no additional requirements as specified in RFC 1035, section 3.3.2.

5.3.3 MB RDATA format

There are no additional requirements as specified in RFC 1035, section 3.3.3.

5.3.4 MD RDATA format

There are no additional requirements as specified in RFC 1035, section 3.3.4 and as amended by RFC 1123, section 6.1.3.6.

5.3.5 MF RDATA format

There are no additional requirements as specified in RFC 1035, section 3.3.5 and as amended by RFC 1123, section 6.1.3.6.

5.3.6 MG RDATA format

There are no additional requirements as specified in RFC 1035, section 3.3.6.

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5.3.7 MINFO RDATA format

There are no additional requirements as specified in RFC 1035, section 3.3.7.

5.3.8 MR RDATA format

There are no additional requirements as specified in RFC 1035, section 3.3.8.

5.3.9 MX RDATA format

There are no additional requirements as specified in RFC 1035, section 3.3.9.

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5.3.10 NULL RDATA format

There are no additional requirements as specified in RFC 1035, section 3.3.10.

5.3.11 NS RDATA format

There are no additional requirements as specified in RFC 1035, section 3.3.11.

5.3.12 PTR RDATA format

There are no additional requirements as specified in RFC 1035, section 3.3.12.

5.3.13 SOA RDATA format

There are no additional requirements as specified in RFC 1035, section 3.3.13.

5.3.14 TXT RDATA format

There are no additional requirements as specified in RFC 1035, section 3.3.14.

5.4 Internet specific RRs

5.4.1 A RDATA format

There are no additional requirements as specified in RFC 1035, section 3.4.1.

5.4.2 WKS RDATA format

There are no additional requirements as specified in RFC 1035, section 3.4.2.

5.5 IN-ADDR.ARPA domain

There are no additional requirements as specified in RFC 1035, section 3.5.

5.6 Messages

5.6.1 Format

5.6.1.1 Header section format

There are no additional requirements as specified in RFC 1035, section 4.1.1 and as amended by RFC 1123: section 6.1.2.3.

5.6.1.2 Question section format

There are no additional requirements as specified in RFC 1035, section 4.1.2.

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5.6.1.3 Resource record format

There are no additional requirements as specified in RFC 1035, section 4.1.3.

5.6.1.4 Message compression

There are no additional requirements as specified in RFC 1035, section 4.1.4 and as amended by RFC 1123, sections 6.1.2.4 and 6.1.3.5.

5.7 Transport

5.7.1 UDP usage

There are no additional requirements as specified in RFC 1035, section 4.2.1 and as amended by RFC 1123, section 6.1.3.2.

5.7.2 TCP usage

There are no additional requirements as specified in RFC 1035, section 4.2.2 and as amended by RFC 1123, section 6.1.3.2.

5.8 Master Files

5.8.1 Format

There are no additional requirements as specified in RFC 1035, section 5.1.

5.8.2 Use of master files to define zones

There are no additional requirements as specified in RFC 1035, section 5.2.

5.9 Inverse queries

There are no additional requirements as specified in RFC 1035, section 6.4.

5.10 Mail Support

There are no additional requirements as specified in RFC 1035, section 8.

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ANNEX A (normative)

DSPICS REQUIREMENTS LIST (DPRL)

A.1 Introduction

This document provides the DOD Standardized Profile Implementation Conformance Statements (DSPICS) Requirements List (DPRL) for implementations of the DOD Standardized Profile (DSP) 2045-17505. The DSPICS for an implementation is generated by completing the DPRL in accordance with the following instructions.

An implementation shall satisfy the mandatory conformance requirements of the base standards referenced in this profile.

An implementation's completed DPRL is called the DSPICS. The DSPICS states which capabilities and options of the protocol have been implemented. The following can use the DSPICS:

(a) the protocol implementor, as a checklist to reduce the risk of failure to conform to the standard through oversight.

(b) the supplier and acquire or potential acquire of the implementation, as a detailed indication of the capabilities of the implementation, stated relative to the common basis for understanding provided by the standard DSPICS proforma.

(c) the user or potential user of the implementation, as a basis for initially checking the possibility of inter-working with another implementation (note that, while inter-working can never be guaranteed, failure to internetwork can often be predicted from incompatible DSPICSs).

(d) a protocol tester, as the basis for selecting appropriate tests against which to assess the claim for conformance of the implementation.

A.1.1 Notation

The following notations and symbols from MIL-HDBK 829, which references ISO/IEC TR 10000-1 and -2, are used in the DPRL to indicate the status of features:

Status Symbols

m	- mandatory
o	- optional
c	- conditional
-	- non-applicable (i.e. logically impossible in the scope of the profile)
x	- excluded or prohibited
i	- out of scope of profile (left as an implementation choice)

Two character combinations may be used for dynamic conformance requirements. In this case, the first character refers to the static (implementation) status, and the second refers to the dynamic (use); thus "mo" means "mandatory to be implemented, optional to be used."

Support Column Symbols

The support of every item as claimed by the implementor is stated by circling the appropriate answer (Yes, No, or N/A) in the support column:

Yes	Supported by the implementation.
No	Not supported by the implementation.

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N/A Not applicable.

A.1.2 Footnotes

Footnotes to the proforma are indicated by superscript numerals. The footnote appears on the page of the first occurrence of the numeral. Subsequent occurrences of a numeral refer to the footnote of the first occurrence.

A.1.3 Instructions for Completing the DPRL

A DSP implementor shows the extent of compliance to a DSP by completing the DPRL; that is, compliance to all mandatory requirements and the options that are not supported are shown. The resulting completed DPRL is called a DSPICS. Where this profile refines the features of the base standards, the requirements expressed in this DPRL shall be applied (as indicated in DPRL items with no "Profile Support" column) to constrain the allowable responses in the base standard DPICS proforma. When this profile makes additional requirements, the "Profile Support" column for such DPRLs shall be completed. In this column, each response shall be selected either from the indicated set of responses, or it shall comprise one or more parameter values as requested. If a conditional requirement is inapplicable, use the Not Applicable (NA) choice. If a mandatory requirement is not satisfied, exception information must be supplied by entering a reference Xi, where i is a unique identifier, to an accompanying rationale for the noncompliance. When the profile requirement is expressed as a two-character combination (as defined in A.1.1 above), the response shall address each element of the requirement; e.g., for the requirement "mo," the possible compliant responses are "yy" or "yn."

A.2 Standards Referenced

This profile specifies the provision of the Domain Names System (DNS) as specified in IAB STD 13 *(RFC 1034: November 1987, Domain Concepts and Facilities, RFC 1035: November 1987, Domain Names - Implementation and Specification) and IAB STD 3 (RFC 1123: October 1989, Requirements For Internet Hosts– Application and Support).*

A.3 DSPICS Requirements List

A.3.1 General Information

A.3.1.1 Implementation Identification

Supplier	
Contact point for queries about the profile	
Implementation name(s) and version(s)	
Date of statement	
Other information: machine name, operating systems, system name	

A.3.1.1.1 Transports

Item	Protocol	Profile	Support
1	Is IAB STD 6, TCP supported ?	o	Yes / No
2	Is IAB STD 7, UDP supported ?	m	Yes

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A.3.1.2 DNS Messages

A.3.1.2.1 Header Section

Item	Fields	Profile		Support		Base Std. References
		Query	Reply	Query	Reply	
1	ID	m	m	Yes	Yes	RFC 1035 sec. 4.1.1 RFC 1123 sec. 6.1.2.3
2	QR	m	m	Yes	Yes	RFC 1035 sec. 4.1.1 RFC 1123 sec. 6.1.2.3
3	OPCODE	m	m	Yes	Yes	RFC 1035 sec. 4.1.1 RFC 1123 sec. 6.1.2.3
3.1	0	m	m	Yes	Yes	RFC 1035 sec. 4.1.1 RFC 1123 sec. 6.1.2.3
3.2	1	m	m	Yes	Yes	RFC 1035 sec. 4.1.1 RFC 1123 sec. 6.1.2.3
3.3	2	m	m	Yes	Yes	RFC 1035 sec. 4.1.1 RFC 1123 sec. 6.1.2.3
3.4	3-15	m	m	Yes	Yes	RFC 1035 sec. 4.1.1 RFC 1123 sec. 6.1.2.3
4	AA	m	m	Yes	Yes	RFC 1035 sec. 4.1.1 RFC 1123 sec. 6.1.2.3
5	TC	m	m	Yes	Yes	RFC 1035 sec. 4.1.1 RFC 1123 sec. 6.1.2.3
6	RD	m	m	Yes	Yes	RFC 1035 sec. 4.1.1 RFC 1123 sec. 6.1.2.3
7	RA	m	m	Yes	Yes	RFC 1035 sec. 4.1.1 RFC 1123 sec. 6.1.2.3
8	Z	m	m	Yes	Yes	RFC 1035 sec. 4.1.1 RFC 1123 sec. 6.1.2.3
9	RCODE	m	m	Yes	Yes	RFC 1035 sec. 4.1.1 RFC 1123 sec. 6.1.2.3
9.1	1	m	m	Yes	Yes	RFC 1035 sec. 4.1.1 RFC 1123 sec. 6.1.2.3
9.2	2	m	m	Yes	Yes	RFC 1035 sec. 4.1.1 RFC 1123 sec. 6.1.2.3
9.3	3	m	m	Yes	Yes	RFC 1035 sec. 4.1.1 RFC 1123 sec. 6.1.2.3
9.4	4	m	m	Yes	Yes	RFC 1035 sec. 4.1.1 RFC 1123 sec. 6.1.2.3
9.5	5	m	m	Yes	Yes	RFC 1035 sec. 4.1.1 RFC 1123 sec. 6.1.2.3
9.6	6-15	m	m	Yes	Yes	RFC 1035 sec. 4.1.1 RFC 1123 sec. 6.1.2.3

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Item	Fields	Profile		Support		Base Std. References
		Query	Reply	Query	Reply	
10	QDCOUNT	m	m	Yes	Yes	RFC 1035 sec. 4.1.1 RFC 1123 sec. 6.1.2.3
11	ANCOUNT	m	m	Yes	Yes	RFC 1035 sec. 4.1.1 RFC 1123 sec. 6.1.2.3
12	NSCOUNT	m	m	Yes	Yes	RFC 1035 sec. 4.1.1 RFC 1123 sec. 6.1.2.3
13	ARCOUNT	m	m	Yes	Yes	RFC 1035 sec. 4.1.1 RFC 1123 sec. 6.1.2.3

A.3.1.2.2 Question Section

Item	Fields	Profile		Support		Base Std. References
		Query	Reply	Query	Reply	
1	QNAME	m	m	Yes	Yes	RFC 1035 sec. 4.1.2
2	QTYPE	m	m	Yes	Yes	RFC 1035 sec. 4.1.2
3	QCLASS	m	m	Yes	Yes	RFC 1035 sec. 4.1.2

A.3.1.2.3 RR Format

Item	Fields	Profile		Support		Base Std. References
		Query	Reply	Query	Reply	
1	NAME	m	m	Yes	Yes	RFC 1035 sec. 3.2.1 RFC 1123 sec. 6.1.2.1
2	TYPE	m	m	Yes	Yes	RFC 1035 sec. 3.2.1 RFC 1123 sec. 6.1.2.1
3	CLASS	m	m	Yes	Yes	RFC 1035 sec. 3.2.1 RFC 1123 sec. 6.1.2.1
4	TTL	m	m	Yes	Yes	RFC 1035 sec. 3.2.1 RFC 1123 sec. 6.1.2.1
5	RDLENGTH	m	m	Yes	Yes	RFC 1035 sec. 3.2.1 RFC 1123 sec. 6.1.2.1
6	RDATA	m	m	Yes	Yes	RFC 1035 sec. 3.2.1 RFC 1123 sec. 6.1.2.1

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A.3.1.2.3.1 RR Type Values

Item	Fields	Profile		Support		Base Std. References
		Query	Reply	Query	Reply	
1	A	m	m	Yes	Yes	RFC 1035 sec. 3.2.2
2	NS	m	m	Yes	Yes	RFC 1035 sec. 3.2.2
3	MD	x	m	Yes	Yes	RFC 1035 sec. 3.2.2
4	MF	x	m	Yes	Yes	RFC 1035 sec. 3.2.2
5	CNAME	m	m	Yes	Yes	RFC 1035 sec. 3.2.2
6	SOA	m	m	Yes	Yes	RFC 1035 sec. 3.2.2
7	MB	m	m	Yes	Yes	RFC 1035 sec. 3.2.2
8	MG	m	m	Yes	Yes	RFC 1035 sec. 3.2.2
9	MR	m	m	Yes	Yes	RFC 1035 sec. 3.2.2
10	NULL	m	m	Yes	Yes	RFC 1035 sec. 3.2.2
11	WKS	m	m	Yes	Yes	RFC 1035 sec. 3.2.2
12	PTR	m	m	Yes	Yes	RFC 1035 sec. 3.2.2
13	HINFO	m	m	Yes	Yes	RFC 1035 sec. 3.2.2
14	MINFO	m	m	Yes	Yes	RFC 1035 sec. 3.2.2
15	MX	m	m	Yes	Yes	RFC 1035 sec. 3.2.2
16	TXT	m	m	Yes	Yes	RFC 1035 sec. 3.2.2

A.3.1.2.3.2 QTYPE Values

Item	Fields	Profile		Support		Base Std. References
		Query	Reply	Query	Reply	
1	AXFR	m	m	Yes	Yes	RFC 1035 sec. 3.2.3
2	MAILB	m	m	Yes	Yes	RFC 1035 sec. 3.2.3
3	MAILA	x	m	Yes	Yes	RFC 1035 sec. 3.2.3
4	*	m	m	Yes	Yes	RFC 1035 sec. 3.2.3

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A.3.1.2.3.3 RR Class Values

Item	Fields	Profile		Support		Base Std. References
		Query	Reply	Query	Reply	
1	IN	m	m	Yes	Yes	RFC 1035 sec. 3.2.4
2	CS	x	m	Yes	Yes	RFC 1035 sec. 3.2.4
3	CH	m	m	Yes	Yes	RFC 1035 sec. 3.2.4
4	HS	m	m	Yes	Yes	RFC 1035 sec. 3.2.4

A.3.1.2.3.4 QCLASS Values

Item	Fields	Profile		Support		Base Std.References
		Query	Reply	Query	Reply	
1	*	m	m	Yes	Yes	RFC 1035 sec. 3.2.5 RFC 1123 sec. 6.1.2.2

A.3.1.2.3.5 RR RDATA Format

Item	Fields	Profile		Support		Base Std. References
		Query	Reply	Query	Reply	
1	CNAME	m	m	Yes	Yes	RFC 1035 sec. 3.3.1
2	HINFO	m	m	Yes	Yes	RFC 1035 sec. 3.3.2
2.1	CPU	m	m	Yes	Yes	RFC 1035 sec. 3.3.2
2.2	OS	m	m	Yes	Yes	RFC 1035 sec. 3.3.2
3	MB	m	m	Yes	Yes	RFC 1035 sec. 3.3.3
4	MD	x	m	Yes	Yes	RFC 1035 sec. 3.3.4 RFC 1123 sec. 6.1.3.6
5	MF	x	m	Yes	Yes	RFC 1035 sec. 3.3.5 RFC 1123 sec. 6.1.3.6
6	MG	m	m	Yes	Yes	RFC 1035 sec. 3.3.6
7	MINFO	m	m	Yes	Yes	RFC 1035 sec. 3.3.7
7.1	RMAILBX	m	m	Yes	Yes	RFC 1035 sec. 3.3.7
7.2	EMAILBX	m	m	Yes	Yes	RFC 1035 sec. 3.3.7
8	MR	m	m	Yes	Yes	RFC 1035 sec. 3.3.8
9	MX	m	m	Yes	Yes	RFC 1035 sec. 3.3.9
9.1	PREFERENCE	m	m	Yes	Yes	RFC 1035 sec. 3.3.9

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Item	Fields	Profile		Support		Base Std. References
		Query	Reply	Query	Reply	
9.2	EXCHANGE	m	m	Yes	Yes	RFC 1035 sec. 3.3.9
10	NULL	m	m	Yes	Yes	RFC 1035 sec. 3.3.10
11	NS	m	m	Yes	Yes	RFC 1035 sec. 3.3.11
12	PTR	m	m	Yes	Yes	RFC 1035 sec. 3.3.12
13	SOA	m	m	Yes	Yes	RFC 1035 sec. 3.3.13
13.1	MNAME	m	m	Yes	Yes	RFC 1035 sec. 3.3.13
13.2	RNAME	m	m	Yes	Yes	RFC 1035 sec. 3.3.13
13.3	SERIAL	m	m	Yes	Yes	RFC 1035 sec. 3.3.13
13.4	REFRESH	m	m	Yes	Yes	RFC 1035 sec. 3.3.13
13.5	RETRY	m	m	Yes	Yes	RFC 1035 sec. 3.3.13
13.6	EXPIRE	m	m	Yes	Yes	RFC 1035 sec. 3.3.13
13.7	MINIMUM	m	m	Yes	Yes	RFC 1035 sec. 3.3.13
14	TXT	m	m	Yes	Yes	RFC 1035 sec. 3.3.14
	INTERNET Specific RRs					
15	A	m	m	Yes	Yes	RFC 1035 sec. 3.4.1
16	WKS	m	m	Yes	Yes	RFC 1035 sec. 3.4.2
16.1	ADDRESS	m	m	Yes	Yes	RFC 1035 sec. 3.4.2
16.2	PROTOCOL	m	m	Yes	Yes	RFC 1035 sec. 3.4.2
16.3	BIT MAP	m	m	Yes	Yes	RFC 1035 sec. 3.4.2

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ANNEX B (informative)

CONCLUDING MATERIAL

B.1 Deviations from the Base Standards/Referenced Profiles

This MIL-STD documents the Domain Name System (DNS) protocol in the ISO/IEC TR 10000, "Framework and Taxonomy of International Standardized Profiles" and MIL-HDBK-829 format. This DSP does not deviate from the protocol as written in the RFC base standards.

The classification of the requirements in RFC 1035 and RFC 1123 have been changed in the DSPICS to the following:

RFC	MIL-STD
MUST	Mandatory
SHOULD	Mandatory
MAY	Optional
SHOULD NOT	Prohibited
MUST NOT	Prohibited

B.2 Subject Term (Keyword) Listing

Directory
Directory Service
DNS
Domain Name
IP
Message
Name Server
Network
Resolver
RFC
RFC 1034
RFC 1035
RFC 1123
TCP
UDP

B.3 Preparing Activity:

Defense Information Systems Agency (DISA) - DC
Project: DCPS-0013

B.4 Reviewing Activity:

Army	PT
Air Force	13, 17, 29, 33
DLA	DH
DMA	MP
DIA	DI
DOT	OST
NSA	NS
OASD	IQ, DO, MA, IR
ODISC4	AC
STRICOM	PT
NAVY	EC, CH, ND, TD
USMC	MC, CG

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-STD 2045-17505	2. DOCUMENT DATE (YYMMDD) 940729

3. DOCUMENT TITLE Information Technology - DOD Standardized Profile - Internet Domain Name System (DNS)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) DSN (If applicable)	7. DATE SUBMITTED (YYMMDD)

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a. NAME Rose D. Satz	b. TELEPHONE (Include Area Code) (1) Commercial 908-532-7732 (2) DSN 992-7732
c. ADDRESS (Include Zip Code) Director JIEO Attn: TBBF Squier Hall Ft. Monmouth, NJ 07703-5613	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 DSN 289-2340

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